

FINAL PROJECT

Frequency Distribution Analysis

Number of Higher Education Institutions by Regency/City in Java and Banten, Indonesia
(2023)

Group Members

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1. Data Collection

1.1 Data Source

The data used in this project is the **number of higher education institutions (universities) per regency/city (kabupaten/kota)** in Indonesia in 2023, published by the Indonesian government open data portal:

Source: data.go.id – “Jumlah Perguruan Tinggi menurut Akreditasi dan Wilayah, Tahun 2023”
<https://data.go.id/dataset/dataset/jumlah-perguruan-tinggi-menurut-akreditasi-2024>

The quantitative variable analyzed is **Total semua**: the total number of higher education institutions (both under Kemendikbudristek and non-Kemendikbudristek) in each regency/city. The observation unit is one regency/city. The dataset covers the provinces of Banten, D.I. Yogyakarta, D.K.I. Jakarta, Jawa Barat, Jawa Tengah, and Jawa Timur, giving **n = 117 observations**, which satisfies the minimum requirement of 100 observations.

1.2 Raw Data (117 observations)

No	Region	Total	No	Region	Total	No	Region	Total	No	Region	Total
1	Kab. Lebak, Banten	8	31	Kab. Pangandaran, Jabar	1	61	Kab. Pati, Jateng	9	91	Kab. Madiun, Jatim	1
2	Kab. Pandeglang, Banten	8	32	Kab. Purwakarta, Jabar	12	62	Kab. Pekalongan, Jateng	2	92	Kab. Magetan, Jatim	2
3	Kab. Serang, Banten	12	33	Kab. Subang, Jabar	11	63	Kab. Pemalang, Jateng	3	93	Kab. Malang, Jatim	15
4	Kab. Tangerang, Banten	28	34	Kab. Sukabumi, Jabar	13	64	Kab. Purbalingga, Jateng	2	94	Kab. Mojokerto, Jatim	14
5	Kota Cilegon, Banten	6	35	Kab. Sumedang, Jabar	8	65	Kab. Purworejo, Jateng	7	95	Kab. Nganjuk, Jatim	11
6	Kota Serang, Banten	13	36	Kab. Tasikmalaya, Jabar	14	66	Kab. Rembang, Jateng	4	96	Kab. Ngawi, Jatim	7
7	Kota Tangerang, Banten	24	37	Kota Bandung, Jabar	78	67	Kab. Semarang, Jateng	11	97	Kab. Pacitan, Jatim	4
8	Kota Tangerang Selatan, Banten	16	38	Kota Banjar, Jabar	6	68	Kab. Sragen, Jateng	2	98	Kab. Pamekasan, Jatim	13
9	Kab. Bantul, DIY	22	39	Kota Bekasi, Jabar	30	69	Kab. Sukoharjo, Jateng	12	99	Kab. Pasuruan, Jatim	15
10	Kab. GunungKidul, DIY	2	40	Kota Bogor, Jabar	14	70	Kab. Tegal, Jateng	8	100	Kab. Ponorogo, Jatim	7
11	Kab. Kulon Progo, DIY	1	41	Kota Cimahi, Jabar	9	71	Kab. Temanggung, Jateng	2	101	Kab. Probolinggo, Jatim	9
12	Kab. Sleman, DIY	38	42	Kota Cirebon, Jabar	10	72	Kab. Wonogiri, Jateng	4	102	Kab. Sampang, Jatim	10
13	Kota Yogyakarta, DIY	29	43	Kota Depok, Jabar	25	73	Kab. Wonosobo, Jateng	2	103	Kab. Sidoarjo, Jatim	11
14	Kota Jakarta Barat, DKI	35	44	Kota Sukabumi, Jabar	9	74	Kota Magelang, Jateng	2	104	Kab. Situbondo, Jatim	8
15	Kota Jakarta Pusat, DKI	54	45	Kota Tasikmalaya, Jabar	15	75	Kota Pekalongan, Jateng	5	105	Kab. Sumenep, Jatim	13
16	Kota Jakarta Selatan, DKI	78	46	Kab. Banjarnegara, Jateng	4	76	Kota Salatiga, Jateng	9	106	Kab. Trenggalek, Jatim	3
17	Kota Jakarta Timur, DKI	78	47	Kab. Banyumas, Jateng	18	77	Kota Semarang, Jateng	47	107	Kab. Tuban, Jatim	8
18	Kota Jakarta Utara, DKI	14	48	Kab. Blora, Jateng	5	78	Kota Surakarta, Jateng	28	108	Kab. Tulungagung, Jatim	11
19	Kab. Bandung, Jabar	17	49	Kab. Boyolali, Jateng	3	79	Kota Tegal, Jateng	3	109	Kota Batu, Jatim	5
20	Kab. Bandung Barat, Jabar	8	50	Kab. Brebes, Jateng	8	80	Kab. Bangkalan, Jatim	11	110	Kota Blitar, Jatim	4
21	Kab. Bekasi, Jabar	26	51	Kab. Cilacap, Jateng	12	81	Kab. Banyuwangi, Jatim	14	111	Kota Kediri, Jatim	16
22	Kab. Bogor, Jabar	48	52	Kab. Demak, Jateng	2	82	Kab. Blitar, Jatim	4	112	Kota Madiun, Jatim	7
23	Kab. Ciamis, Jabar	9	53	Kab. Grobogan, Jateng	4	83	Kab. Bojonegoro, Jatim	13	113	Kota Malang, Jatim	51
24	Kab. Cianjur, Jabar	18	54	Kab. Jepara, Jateng	2	84	Kab. Bondowoso, Jatim	9	114	Kota Mojokerto, Jatim	4
25	Kab. Cirebon, Jabar	23	55	Kab. Karanganyar, Jateng	8	85	Kab. Gresik, Jatim	15	115	Kota Pasuruan, Jatim	3
26	Kab. Garut, Jabar	19	56	Kab. Kebumen, Jateng	6	86	Kab. Jember, Jatim	20	116	Kota Probolinggo, Jatim	1
27	Kab. Indramayu, Jabar	16	57	Kab. Kendal, Jateng	4	87	Kab. Jombang, Jatim	15	117	Kota Surabaya, Jatim	85
28	Kab. Karawang, Jabar	12	58	Kab. Klaten, Jateng	5	88	Kab. Kediri, Jatim	10			
29	Kab. Kuningan, Jabar	8	59	Kab. Kudus, Jateng	4	89	Kab. Lamongan, Jatim	12			
30	Kab. Majalengka, Jabar	6	60	Kab. Magelang, Jateng	5	90	Kab. Lumajang, Jatim	11			

2. Frequency Distribution Table

2.1 Constructing the Table (Sturges' Method)

Step 1. Highest value = **85** (Kota Surabaya), lowest value = **1**. Range = $85 - 1 = 84$.

Step 2. Number of class intervals using Sturges' formula:

$$k = 1 + 3.322 \log n = 1 + 3.322 \log 117 = 1 + 3.322 (2.0682) = 7.8705 \approx 8 \text{ classes (rounded up)}$$

Step 3. Width of class interval:

$$c = (\text{highest} - \text{lowest}) / k = (85 - 1) / 8 = 10.5 \approx 11 \text{ (rounded up so all data fits)}$$

Step 4. The lowest class limit is the lowest value, 1, so the first class is 1 – 11 with class limits 0.5 – 11.5. With 8 classes of width 11, the last class is 78 – 88 (77.5 – 88.5), which covers the maximum value 85. All 117 observations are accommodated.

2.2 Frequency Distribution Table

Class Interval	Class Limits	Middle Value	Frequency
1 – 11	0.5 – 11.5	6	69
12 – 22	11.5 – 22.5	17	30
23 – 33	22.5 – 33.5	28	8
34 – 44	33.5 – 44.5	39	2
45 – 55	44.5 – 55.5	50	4
56 – 66	55.5 – 66.5	61	0
67 – 77	66.5 – 77.5	72	0
78 – 88	77.5 – 88.5	83	4
Total			117

2.3 Relative Frequency Table

Class Interval	Frequency	Relative Frequency	Percentage
1 – 11	69	69/117	58.97%
12 – 22	30	30/117	25.64%
23 – 33	8	8/117	6.84%
34 – 44	2	2/117	1.71%
45 – 55	4	4/117	3.42%
56 – 66	0	0/117	0.00%
67 – 77	0	0/117	0.00%
78 – 88	4	4/117	3.42%
Total	117		100.00%

2.4 Cumulative Frequency Table

Class Interval	Frequency	Less Than	Cumulative Frequency	Cumulative %
1 – 11	69	≤ 11.5	69	58.97%
12 – 22	30	≤ 22.5	99	84.62%
23 – 33	8	≤ 33.5	107	91.45%
34 – 44	2	≤ 44.5	109	93.16%
45 – 55	4	≤ 55.5	113	96.58%
56 – 66	0	≤ 66.5	113	96.58%
67 – 77	0	≤ 77.5	113	96.58%
78 – 88	4	≤ 88.5	117	100.00%

3. Data Visualization

3.1 Histogram

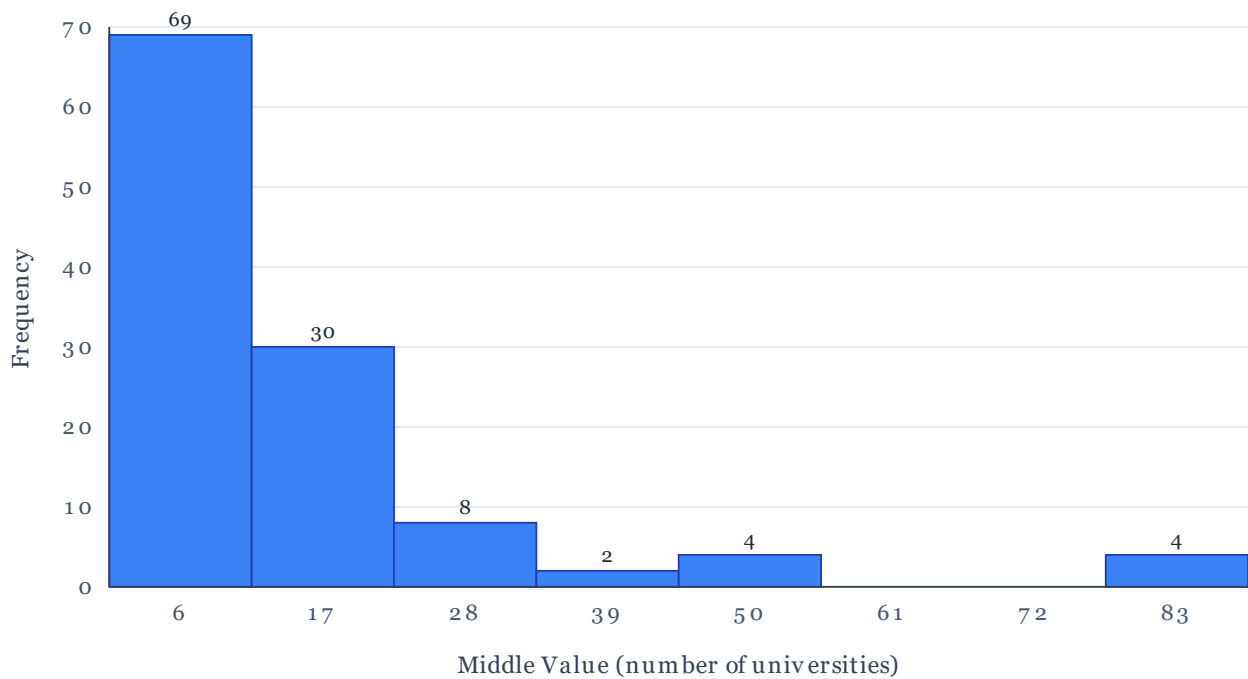


Figure 1. Histogram of the number of universities per regency /city

3.2 Frequency Polygon

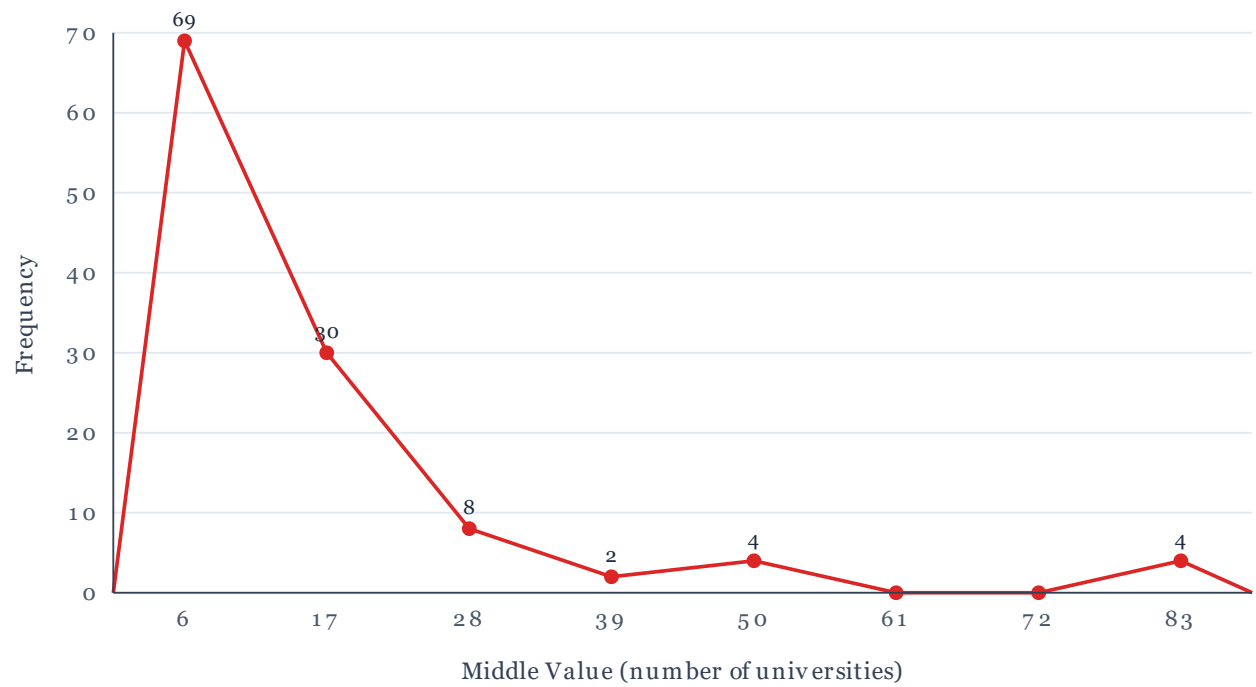


Figure 2. Frequency polygon (line through class middle values)

3.3 Ogive (Cumulative Frequency Curve)



Figure 3. Less-than ogive plotted at upper class limits

4. Descriptive Analysis

4.1 Minimum, Maximum, and Range

- **Minimum** = 1 university (Kab. Kulon Progo, Kab. Pangandaran, Kab. Madiun, Kota Probolinggo).
- **Maximum** = 85 universities (Kota Surabaya), followed by Kota Bandung, Kota Jakarta Selatan, and Kota Jakarta Timur with 78 each.
- **Range** = $85 - 1 = 84$, a very wide spread relative to typical values.

4.2 Distribution Shape

The distribution is **strongly right-skewed (positively skewed)**. The histogram shows the tallest bar at the first class (1 – 11) with 69 regencies/cities (58.97% of all observations), frequencies falling off rapidly, and a long thin tail stretching to the right. Two classes in the tail (56 – 66 and 67 – 77) are even empty before a small cluster of 4 extreme values appears in the last class (78 – 88).

The ogive confirms this: it rises very steeply at the beginning and flattens quickly — 99 of 117 regions (84.62%) already have 22 or fewer universities, and 107 regions (91.45%) have 33 or fewer.

4.3 Key Insights

1. **Higher education is highly concentrated.** The majority of regencies/cities (58.97%) host at most 11 institutions, while a handful of large cities host dozens.
2. **The extreme values are all major metropolitan cities:** Surabaya (85), Bandung (78), Jakarta Selatan (78), and Jakarta Timur (78). These are outliers far from the rest of the data, creating the isolated last class in the histogram.
3. **Kabupaten vs Kota gap.** Nearly all high-frequency regions in the upper classes are cities (kota), while regencies (kabupaten) dominate the lowest class, showing that university access is centered in urban areas.
4. **Consequence of skewness.** Because of the strong right skew, the mean would be pulled well above the median; the median or mode is a more representative measure of a “typical” regency/city than the mean for this data.

4.4 Conclusion

Using Sturges’ formula, the 117 observations were grouped into 8 classes of width 11. The frequency distribution, histogram, frequency polygon, and ogive all show a strongly right-skewed distribution: most regencies/cities in Java and Banten have very few higher education institutions, while a small number of metropolitan cities concentrate most of them. This reflects the real-world centralization of higher education in Indonesia’s big cities.